

Failure of rational injectivity for maximal coarse assembly

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Abstract

We construct a uniformly discrete bounded-geometry space whose maximal coarse assembly map has an infinite-order element in its kernel. The space is a coarse disjoint union of finite connected graphs of uniformly bounded degree, so maximal coarse assembly need not be rationally injective even for such graph unions.

1 Introduction

Coarse assembly relates the topology visible at bounded scales in a metric space to the indices of operators controlled by its large-scale geometry. Its rational injectivity asks whether a non-torsion topological class can have zero coarse index. We prove that this can occur even when the analytic target is the maximal Roe algebra and the space is a coarse disjoint union of finite graphs of bounded degree.

Let Z be a uniformly discrete proper metric space of bounded geometry; thus every ball of a fixed radius contains a uniformly bounded number of points. Write $P_r(Z)$ for the Rips complex whose simplices are the finite subsets of diameter at most r , and set

$$\mathrm{KX}_q(Z) = \varinjlim_{r \rightarrow \infty} \mathrm{KK}_q(C_0(P_r(Z)), \mathbb{C}).$$

All K-theory and K-homology in this article are complex. On the ample module $H_Z = \ell^2(Z) \otimes \ell^2(\mathbb{N})$, let $\mathbb{C}[Z]$ be the algebra of bounded, locally compact, finite-propagation operators. Local compactness means that aT and Ta are compact for every $a \in C_0(Z)$, and finite propagation means that $T_{xy} = 0$ whenever $d(x, y)$ exceeds some fixed bound. In particular, the compact matrix entries of T need not satisfy a uniform rank bound. The universal C^* -completion is the *maximal Roe algebra* $C_{\max}^*(Z)$. The canonical maximal coarse assembly map is

$$\mu_Z^{\max} : \mathrm{KX}_q(Z) \longrightarrow K_q(C_{\max}^*(Z)).$$

We recall the needed norm estimates and the index construction in Section 4.

A *coarse disjoint union* of finite graphs is a metric on their disjoint union which restricts to the graph path metric on each component and for which only finitely many pairs of distinct components are within any fixed distance. We take such metrics to be uniformly discrete.

Theorem 1.1. *There exist finite connected graphs $(X_j)_{j \geq 1}$ with a common finite bound on vertex degree, a coarse disjoint union $X = \bigsqcup_{j \geq 1} X_j$, and an infinite-order class $\alpha \in \mathrm{KX}_1(X)$ such that*

$$\mu_X^{\max}(\alpha) = 0.$$

Consequently, $\mu_X^{\max} \otimes \mathbb{Q}$ is not injective.

The theorem strengthens the reduced coarse-assembly result proved in the companion paper [13]. If $C^*(X)$ denotes the operator-norm closure of $\mathbb{C}[X]$, the canonical quotient $C_{\max}^*(X) \rightarrow C^*(X)$ carries the maximal index to the reduced index. Our class α therefore also lies in the reduced assembly kernel. The stronger conclusion requires a proof in the universal norm: vanishing of a reduced index alone does not imply vanishing of its maximal lift.

1.1 Context and prior work

Roe developed coarse cohomology and coarse index theory as tools for large-scale geometry and index-theoretic questions on noncompact manifolds [15]. Higson and Roe formulated and studied the coarse Baum–Connes conjecture in this framework [10]. The connection with Novikov-type questions is that injectivity retains the topological information carried by higher indices. The present theorem concerns the non-equivariant coarse assembly map displayed above.

Several fundamental positive results relate assembly to geometric control. Yu proved coarse assembly isomorphism for bounded-geometry spaces of finite asymptotic dimension [20, Theorem 7.1] and established the coarse Baum–Connes conjecture for bounded-geometry spaces admitting a uniform embedding into Hilbert space [21]. Skandalis, Tu, and Yu developed the coarse-groupoid approach and its relation to these embedding methods [17]. These results explain the importance of geometric hypotheses when seeking injectivity or isomorphism theorems.

Earlier failures of injectivity also show why bounded geometry and the domain of assembly matter. Yu constructed a coarse-assembly kernel class for a space without bounded geometry [20, Section 8, Proposition 8.1]. Dranishnikov, Ferry, and Weinberger constructed integral index-kernel examples with locally finite K-homology as domain; their kernel classes disappear under coarsening [6, Section 6, Remark 6.2]. Here the class belongs to the actual Rips colimit for a bounded-geometry graph union and remains nonzero after rationalization.

Finite quotients and expanding graphs also exposed limitations of general assembly assertions. Higson, Lafforgue, and Skandalis constructed counterexamples in group and groupoid settings and proved reduced coarse nonsurjectivity for certain unions of congruence Cayley graphs [9, Section 6]. Nonsurjectivity is a different obstruction from the kernel in Theorem 1.1. Their separate Hausdorff-groupoid kernel construction combines an odd Bott class with a Kazhdan projection: a nonzero full-assembly image becomes zero after reduction [9, Section 5]. This is a methodological predecessor, but its behavior also makes it essential to distinguish the two analytic targets.

The maximal Roe algebra and its assembly map were developed in work of Gong, Wang, and Yu [7] and Oyono-Oyono and Yu [14]; the latter analyzed maximal assembly for specific families of expanders arising from finite group quotients. Willett and Yu further studied higher indices for expanding and large-girth graph families [19]. Špakula and Willett proved that the maximal-to-reduced comparison induces a K-theory isomorphism for spaces admitting a coarse embedding into Hilbert space [18]. Chen, Wang, and Yu proved the maximal coarse Baum–Connes conjecture under the more general hypothesis of a fibred coarse embedding into Hilbert space [4]. More recent injectivity results concern finite-group extensions whose kernel and quotient families admit such fibred embeddings [5]; maximal isomorphism results also hold for coarse fibrations with uniformly property-A fibers, a fibred-coarsely-embeddable base, and uniform coarse equivalences between fibers and preimages of fixed-radius balls [8]. Finite fibers alone do not supply these uniform hypotheses. Theorem 1.1 shows that rational injectivity cannot hold for all bounded-geometry graph unions without further assumptions.

Our analytic tools belong to this established maximal-index framework. We give the norm and functoriality arguments needed here, with particular attention to the points where the universal and operator norms must be distinguished; see also [14, 16, 18]. Kazhdan’s property (T) [2, 12] supplies averaging in the full group C*-algebra. The topological part uses integral line-bundle classification, the K-theory of projective spaces, and the standard K-homology pairing [1, 3, 11].

1.2 The construction and the proof

The base graphs are three-dimensional discrete tori B_j whose side lengths tend to infinity. Over each B_j we place a finite quotient of $\mathrm{SL}_3(\mathbb{Z})$, obtaining a graph Y_j with Cayley edges inside the fibers. Transport around the first two base circles is given by commuting elements of that quotient. These elements cease to commute at a higher congruence level. The graph X_j joins two copies of B_j through Y_j : its middle has a long product collar, and toward either end the fiber quotients are reduced one level at a time until they become trivial. This keeps all vertex degrees bounded. Section 2 gives the construction.

Write $B = \bigsqcup_j B_j$, $Y = \bigsqcup_j Y_j$, and let

$$f : Y \longrightarrow B, \quad i_{\pm} : B \longrightarrow X$$

be the middle projection and the two end inclusions. A torus K-homology class on each component defines a single class $\beta \in \mathrm{KX}_1(B)$. The class of interest is

$$\alpha = (i_+)_* \beta - (i_-)_* \beta.$$

Its persistence and its zero index come from different features of the middle projection.

For persistence, take a line bundle $\mathcal{L}$ on each continuous base torus, with first Chern class the product of the first two circle classes. Its reduced K-theory class, multiplied by the third circle’s odd class, pairs to one with the chosen K-homology class. At every fixed Rips scale, the lifted commutator calculation shows that the pullback of $\mathcal{L}$ to the middle admits a 2^j -th root for all sufficiently large j . A uniform dimension bound for the Rips complexes converts these roots into divisibility by a growing power of two in integral K-theory. Mayer–Vietoris then prevents any fixed nonzero multiple of the endpoint difference from vanishing. Section 3 makes this argument before taking either the scale colimit or rational coefficients.

For the index, projection from the middle induces a surjection

$$f_* : K_1(C_{\max}^*(Y)) \longrightarrow K_1(C_{\max}^*(B)).$$

To construct a splitting, we transport each base matrix entry horizontally between the fibers and then apply the Kazhdan projection onto the constant vectors in each fiber. The transport lift need not preserve products; averaging makes it multiplicative, and projecting the resulting homomorphism back to the base induces the identity on K-theory. Section 5 proves these identities in the maximal completion using two controls from Section 4: agreement of maximal and operator norms on finite vertex supports, and uniform norm estimates at fixed propagation.

Finally, Section 6 shows that the two composites $(i_+)_* f_*$ and $(i_-)_* f_*$ agree on analytic K_1 . The end maps are not uniformly close: their separation grows with j . Instead, we place orthogonal copies of an operator at every height and rotate these copies simultaneously by one vertical step. After cancelling the common interior copies, the endpoint maps agree modulo compact operators, and injectivity of $K_1(C_{\max}^*(X)) \rightarrow K_1(C_{\max}^*(X)/\mathcal{K})$ removes that quotient. Surjectivity of f_* and naturality of assembly then give the zero index.

2 Congruence fibers over discrete tori

We construct a graph with two torus ends and a long central region of constant fiber size. The fiber quotient changes one congruence level at a time outside this region. These gradual changes keep the degrees bounded, while the long central region will isolate the topological obstruction at any fixed Rips scale.

2.1 The finite quotients and their twists

Fix a finite symmetric generating set S of $\Gamma = \mathrm{SL}_3(\mathbb{Z})$. We use Kazhdan's theorem that Γ has property (T) [2, 12]. For $k \geq 1$, let Q_k be the image of Γ under entrywise reduction modulo 2^k , and put $Q_0 = \{1\}$. The natural reduction maps will always be denoted by their source and target.

Lemma 2.1. *The maps $Q_{k+1} \rightarrow Q_k$ are surjective and have kernels of size at most 2^9 . For every fixed word radius R , the reduction $\Gamma \rightarrow Q_k$ is injective on the S -word ball of radius R for all sufficiently large k .*

Proof. Surjectivity follows from the definition as images of the same group. For $k \geq 1$, each matrix modulo 2^k has at most 2^9 entrywise lifts modulo 2^{k+1} . For $k = 0$, the same bound follows by counting all binary 3×3 matrices. Finally, a word ball is a finite set of integral matrices. If two of its elements differ, some difference of entries is a nonzero integer and is not divisible by arbitrarily large powers of two. Taking the maximum of the finitely many required levels proves injectivity. $\square$

For each $j \geq 1$, set

$$m_j = 2^j, \quad l_j = 100(j+1), \quad U_j = 1 + m_j E_{12}, \quad V_j = 1 + m_j E_{23}, \quad (2.1)$$

where E_{ab} denotes a matrix unit. A direct multiplication gives

$$U_j V_j U_j^{-1} V_j^{-1} = 1 + m_j^2 E_{13}. \quad (2.2)$$

Thus U_j and V_j commute in Q_{2j} and in every lower quotient, although their commutator remains visible in Q_{3j} . This difference between the two levels is the source of the divisibility argument in Section 3.

Let $B_j = (\mathbb{Z}/l_j\mathbb{Z})^3$ with its axis-neighbor graph. Represent each coordinate by an integer in $\{0, \dots, l_j - 1\}$. An oriented edge has signed seam numbers $(a, b, c) \in \mathbb{Z}^3$: a positive step from $l_j - 1$ to 0 contributes 1 in that coordinate, a reverse step contributes -1 , and all other contributions vanish. Equivalently, the unit displacement along the edge is the difference of the chosen representatives plus $l_j(a, b, c)$. For $0 \leq k \leq 2j$, define

$$g_{j,k}(a, b) = V_j^b U_j^a \pmod{2^k} \in Q_k, \quad (2.3)$$

with the evident interpretation in Q_0 . Only the first two circle coordinates will twist the fibers.

The graph $Y_{j,k}$ has vertex set $B_j \times Q_k$. Its edges are of two types:

- over a fixed $u \in B_j$, connect (u, q) to (u, sq) for $s \in S$;
- for an oriented base edge $u \rightarrow v$ with seam numbers (a, b, c) , connect (u, q) to $(v, qg_{j,k}(a, b))$.

The first action is on the left and the second on the right, so they commute. Reversing a base edge reverses its matching, because U_j and V_j commute in Q_k . We discard loops and repeated edges. Each fiber Cayley graph is connected, and the horizontal matchings connect the fibers over the connected graph B_j . Hence $Y_{j,k}$ is connected. Write

$$Y_j = Y_{j,2j}.$$

The metric on Y_j is its intrinsic graph metric.

2.2 The two ends and the central collar

For each integer t with $-3j \leq t \leq 3j$, put

$$k_j(t) = \begin{cases} 2j, & |t| \leq j, \\ 3j - |t|, & j \leq |t| \leq 3j. \end{cases} \quad (2.4)$$

At height t , place a copy of $Y_{j,k_j(t)}$. Connect successive slices over the same base vertex by the reduction map from the larger quotient to the smaller one, or by the identity when the levels agree. The resulting finite graph is X_j .

At height $\pm 3j$ the quotient is trivial, so each end is a copy of B_j . The induced graph on the heights $-j, \dots, j$ is the product of Y_j with the interval graph: its horizontal edges are those of Y_j , and its vertical edges preserve both u and q . We call this subgraph the *central collar*. Paths in X_j may take shortcuts outside the collar; whenever we use its product structure, we will first show that the paths under consideration cannot leave it.

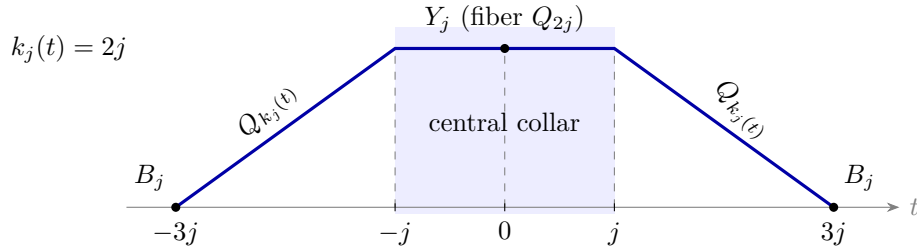


Figure 1: The congruence level as a function of height in X_j . Every slice lies over the same discrete torus B_j . Outside the central collar, each vertical step changes the quotient level by one. The drawing records the level profile, not a geometric embedding of the graph or its fiber cardinalities.

The gradual reductions in Figure 1 have bounded fibers by Lemma 2.1. They therefore preserve a uniform degree bound even though the middle quotient grows rapidly with j .

Lemma 2.2. *The graphs B_j, Y_j, X_j are finite and connected, with vertex degrees bounded independently of j . Their projections to B_j are 1-Lipschitz. Each end copy of B_j is isometrically embedded in X_j .*

Proof. Only the claims about X_j remain to be checked. Consecutive slices are connected by surjective reduction maps, so X_j is connected. A vertex has at most $|S|$ fiber neighbors, six base neighbors, and at most 2^9 vertical neighbors on either side. Thus $|S| + 6 + 2^{10}$ is a common bound. Projection sends an edge to a base edge or a point, proving the Lipschitz assertion. Within an end, every base path is available. Projection gives the opposite distance inequality, so the end metric is exactly the metric of B_j . $\square$

2.3 Coarse disjoint unions

We now fix metrics on

$$B = \bigsqcup_{j \geq 1} B_j, \quad Y = \bigsqcup_{j \geq 1} Y_j, \quad X = \bigsqcup_{j \geq 1} X_j.$$

In B_j choose the zero vertex, in Y_j the vertex $(0, 1)$, and in X_j the vertex $(0, 1)$ at height zero. In each union, join the chosen vertices of components j and $j + 1$ by an auxiliary edge of length $100 + j^2$, and use the resulting path distances on the original vertex set. These edges define the metric; they add no vertices to the space.

A path which leaves and returns to the same component must pass through its chosen vertex both times. Consequently its use of the auxiliary edges cannot shorten an intrinsic component distance. At each fixed distance bound, cross-component pairs occur in only finitely many finite components. The resulting metrics are uniformly discrete and proper. They have bounded geometry because the weighted graphs have uniformly bounded degrees and all edge lengths are at least one: a ball of radius R contains no more vertices than a ball of graph radius $\lfloor R \rfloor$ in a graph of the same degree bound.

Projection and end inclusion define maps

$$f : Y \longrightarrow B, \quad i_{\pm} : B \longrightarrow X. \quad (2.5)$$

On each component they are 1-Lipschitz. For a fixed input distance bound, the remaining pairs lie in a finite set, so their image distances also have a finite bound. Thus the maps are bornologous. Preimages of bounded sets are finite, so the maps are proper. At scale 3 none of the Rips complexes has a cross-component simplex. In particular, the end maps induce proper simplicial inclusions

$$i_{\pm, 3} : P_3(B) \longrightarrow P_3(X).$$

These are the maps used to form the K-homology class. The next section shows why its two endpoint contributions remain distinct at every fixed scale.

3 An integral class that persists at every scale

We construct a class $\beta \in KX_1(B)$ whose images at the two ends of X have an infinite-order difference. The obstruction to identifying these images will be integral: at any fixed Rips scale, an appropriate K -cohomology class becomes divisible by larger and larger powers of two over the middle graphs Y_j .

All K -groups in this section are complex and integral. For a finite complex M , we write $K^*(M)$ for topological K -theory and $K_*(M) = KK_*(C(M), \mathbb{C})$ for analytic K -homology. The pairing

$$\langle \cdot, \cdot \rangle : K^1(M) \times K_1(M) \longrightarrow \mathbb{Z}$$

is the Kasparov index pairing. In particular, for a continuous map $h : M \rightarrow N$ between finite complexes,

$$\langle \eta, h_* \xi \rangle = \langle h^* \eta, \xi \rangle. \quad (3.1)$$

We use the usual Mayer–Vietoris sequence for a finite complex covered by two subcomplexes; see [11] for these K -homology constructions.

3.1 The classes on the two ends

Let

$$T_j = (\mathbb{R}/l_j\mathbb{Z})^3.$$

Give T_j its unit cubical subdivision and triangulate the cubes periodically. The vertices are B_j , and the vertices of every cube have graph distance at most 3. The resulting triangulated torus is therefore a subcomplex of $P_3(B_j)$; write $\kappa_j : T_j \hookrightarrow P_3(B_j)$ for its inclusion.

Let $u_1, u_2, u_3 \in H^1(T_j; \mathbb{Z})$ be the positively oriented coordinate classes. Choose a line bundle $\mathcal{L}_j$ pulled back from the first two circles with

$$c_1(\mathcal{L}_j) = u_1 \smile u_2,$$

and let $e_3 \in K^1(T_j)$ be the pullback of the positive generator of $K^1(S^1)$ under projection to the third circle. Define

$$\theta_j = ([\mathcal{L}_j] - 1)e_3 \in K^1(T_j). \quad (3.2)$$

This class is primitive. Indeed, the exterior-algebra description of $K^*((S^1)^3)$ identifies $[\mathcal{L}_j] - 1$ with a generator of the reduced K^0 of the first two circles, and θ_j with the product of the three circle generators. Here the first assertion also follows from the first Chern class, which identifies $\widetilde{K}^0((S^1)^2)$ with $H^2((S^1)^2; \mathbb{Z})$. The Künneth and universal coefficient theorems therefore give a class

$$\tau_j \in K_1(T_j), \quad \langle \theta_j, \tau_j \rangle = 1. \quad (3.3)$$

More explicitly, $K^*(T_j)$ is free abelian, so the universal coefficient theorem identifies $K_1(T_j)$ with $\text{Hom}(K^1(T_j), \mathbb{Z})$, and a primitive element admits an integer-valued functional taking value 1 on it; see [3, Theorems 23.1.1 and 23.1.3].

The component separation gives $P_3(B) = \bigsqcup_j P_3(B_j)$. Countable additivity in the first variable of KK gives

$$\text{KK}_1(C_0(P_3(B)), \mathbb{C}) \cong \prod_{j \geq 1} K_1(P_3(B_j)). \quad (3.4)$$

This is a product, so the entire sequence $((\kappa_j)_* \tau_j)_j$ determines a class β_3 in the group on the left; see [3, Theorem 19.7.1]. Write β for its image in $\text{KX}_1(B)$ and set

$$\alpha_3 = (i_{+,3})_* \beta_3 - (i_{-,3})_* \beta_3, \quad \alpha = [\alpha_3] \in \text{KX}_1(X). \quad (3.5)$$

Thus $\alpha = (i_+)_* \beta - (i_-)_* \beta$. We will prove that every nonzero integer multiple of α_3 remains nonzero at every later Rips scale.

3.2 Retractions to the tori

Fix an integer $r \geq 3$. From now on, thresholds on j may depend on r , but not on a simplex or a vertex of the corresponding component.

Lemma 3.1. *If $l_j > 4r$, there is a continuous map*

$$p_{B,j} : P_r(B_j) \longrightarrow T_j$$

whose restriction to $T_j \subseteq P_3(B_j) \subseteq P_r(B_j)$ is the identity. The graph projections define continuous maps

$$p_{Y,j} : P_r(Y_j) \longrightarrow T_j, \quad p_{X,j} : P_r(X_j) \longrightarrow T_j$$

by composition with $p_{B,j}$.

Proof. For vertices u, v of a Rips simplex, each coordinate difference has a unique representative $\delta_i(u, v) \in \mathbb{Z}$ with $|\delta_i(u, v)| \leq r$. On a triangle (u, v, w) , the integer

$$\delta_i(u, v) + \delta_i(v, w) - \delta_i(u, w)$$

is a multiple of l_j and has absolute value at most $3r < l_j$; it therefore vanishes. Consequently the vertices of each simplex admit lifts to $\mathbb{Z}^3$ realizing all these differences. The lifts are unique up to a common element of $l_j\mathbb{Z}^3$. Affine interpolation, followed by reduction modulo $l_j\mathbb{Z}^3$, gives a map on the simplex independent of that choice. The maps agree on faces and define $p_{B,j}$.

On each simplex of the triangulated torus, this interpolation is its original parametrization, so the restriction is the identity. The projections $Y_j \rightarrow B_j$ and $X_j \rightarrow B_j$ are 1-Lipschitz by Lemma 2.2; hence they induce simplicial maps of the corresponding Rips complexes. $\square$

The maps in Lemma 3.1 allow us to compare the same torus class at the ends and in the middle. At either end its pairing with τ_j is 1. In the middle, a congruence calculation forces divisibility.

3.3 The congruence calculation

Lemma 3.2. *For every fixed integer $r \geq 3$ and all sufficiently large j ,*

$$p_{Y,j}^*(u_1 \smile u_2) \in 2^j H^2(P_r(Y_j); \mathbb{Z}). \quad (3.6)$$

Equivalently, there is a line bundle $\mathcal{L}'_{j,r}$ on $P_r(Y_j)$ such that

$$p_{Y,j}^* \mathcal{L}_j \cong (\mathcal{L}'_{j,r})^{\otimes 2^j}. \quad (3.7)$$

Proof. Take j large enough that $l_j > 4r$ and reduction $\Gamma \rightarrow Q_{2j}$ is injective on the word ball of radius $2r$; the latter condition is available from Lemma 2.1. In this proof write $m = 2^j$, suppress j in U_j, V_j , and abbreviate $g_{j,k}$ to g_k .

We first describe integral cocycles representing $p_{Y,j}^* u_1, p_{Y,j}^* u_2$. For an oriented Rips edge (x, y) , let $v_x, v_y \in B_j$ be its projected endpoints and choose their coordinate representatives in $\{0, \dots, l_j - 1\}$. Define its first two seam numbers $a_{xy}, b_{xy} \in \mathbb{Z}$ by

$$\delta_1(v_x, v_y) = (v_y)_1 - (v_x)_1 + l_j a_{xy}, \quad \delta_2(v_x, v_y) = (v_y)_2 - (v_x)_2 + l_j b_{xy}.$$

The triangle identities in the proof of Lemma 3.1 show that a, b are cocycles. The seam numbers record the winding of the interpolated coordinate maps, so $[a] = p_{Y,j}^* u_1$ and $[b] = p_{Y,j}^* u_2$.

To prove divisibility by m , it suffices to show that $a \smile b$ becomes a coboundary modulo m . We will construct a $\mathbb{Z}/m\mathbb{Z}$ -valued 1-cochain c with $\delta c = -a \smile b$ by comparing edge transports after lifting from Q_{2j} to Q_{3j} .

Order the vertices of $P_r(Y_j)$. For every edge $x < y$, choose a graph path from x to y of length at most r , and write $x = (v_x, q_x)$ and $y = (v_y, q_y)$. The left Cayley steps along this path give a word $w_{xy} \in \Gamma$ of length at most r . Left multiplications commute with all horizontal right multiplications. Moreover, U, V commute modulo m^2 , and the total seam numbers of the projected path are a_{xy}, b_{xy} : its coordinate displacement is bounded by r and therefore is the unique small lift used above. It follows that

$$q_y = \overline{w}_{xy} q_x g_{2j}(a_{xy}, b_{xy}) \quad \text{in } Q_{2j}. \quad (3.8)$$

Here a bar denotes reduction modulo m^2 .

On an ordered triangle with vertices $x_0 < x_1 < x_2$, cocycle additivity and commutation modulo m^2 give

$$g_{2j}(a_{01}, b_{01})g_{2j}(a_{12}, b_{12}) = g_{2j}(a_{02}, b_{02}).$$

Substitution in (3.8) and cancellation give $\bar{w}_{02} = \bar{w}_{12}\bar{w}_{01}$. Both w_{02} and $w_{12}w_{01}$ lie in the word ball of radius $2r$, so our injectivity assumption yields the equality in Γ :

$$w_{02} = w_{12}w_{01}. \quad (3.9)$$

We now lift this calculation from modulus m^2 to modulus m^3 . Every element of

$$N = \ker(Q_{3j} \longrightarrow Q_{2j})$$

has the form $1 + m^2D$ modulo m^3 , with D determined modulo m . Multiplication adds the matrices D , since m^4 is divisible by m^3 . Thus N is abelian and

$$d : N \longrightarrow \mathbb{Z}/m\mathbb{Z}, \quad d(1 + m^2D) = D_{13} \pmod{m}$$

is a homomorphism. Every element of N commutes with every matrix that is 1 modulo m , in particular with

$$\tilde{g}(a, b) = V^b U^a \pmod{m^3}.$$

Direct multiplication of the elementary matrices gives

$$\begin{aligned} C &= UVU^{-1}V^{-1} = 1 + m^2E_{13} \pmod{m^3}, & d(C) &= 1, \\ \tilde{g}(a, b)\tilde{g}(a', b') &= \tilde{g}(a + a', b + b')C^{ab'}. \end{aligned} \quad (3.10)$$

Choose a lift $\tilde{q}_x \in Q_{3j}$ of each q_x , and denote the reduction of $w \in \Gamma$ to Q_{3j} by $\tilde{w}$. Equation (3.8) defines $n_{xy} \in N$ by

$$\tilde{q}_y = \tilde{w}_{xy}\tilde{q}_x\tilde{g}(a_{xy}, b_{xy})n_{xy}. \quad (3.11)$$

For an ordered triangle, abbreviate $\tilde{g}(a_{ab}, b_{ab})$ to $\tilde{g}_{ab}$. Substituting twice in (3.11) gives

$$\begin{aligned} \tilde{q}_2 &= \tilde{w}_{12}\tilde{w}_{01}\tilde{q}_0\tilde{g}_{01}n_{01}\tilde{g}_{12}n_{12} \\ &= \tilde{w}_{02}\tilde{q}_0\tilde{g}_{01}\tilde{g}_{12}n_{01}n_{12} \\ &= \tilde{w}_{02}\tilde{q}_0\tilde{g}_{02}C^{a_{01}b_{12}}n_{01}n_{12}. \end{aligned}$$

The second equality moves n_{01} only past $\tilde{g}_{12}$, which it commutes with; it uses no commutation with $\tilde{q}_0$ or with a Cayley word. Comparing with the direct edge 02 therefore gives

$$n_{02} = C^{a_{01}b_{12}}n_{01}n_{12}.$$

Apply d and put $c_{xy} = d(n_{xy})$. With the usual simplicial coboundary convention, this identity becomes

$$(a \smile b)_{012} = a_{01}b_{12} = c_{02} - c_{01} - c_{12} = -(\delta c)_{012} \quad \text{in } \mathbb{Z}/m\mathbb{Z}.$$

Hence the reduction of $[a \smile b]$ modulo m is zero. The cohomology sequence of

$$0 \longrightarrow \mathbb{Z} \xrightarrow{m} \mathbb{Z} \longrightarrow \mathbb{Z}/m\mathbb{Z} \longrightarrow 0$$

identifies the kernel of reduction on H^2 with the image of multiplication by m . This proves (3.6), without a torsion-freeness assumption on $H^2(P_r(Y_j); \mathbb{Z})$. Finally, line bundles on a finite complex are classified by their integral first Chern class, and the first Chern class of a tensor power is the corresponding integer multiple. This proves (3.7). $\square$

3.4 From line-bundle roots to divisibility in K^1

For fixed r , bounded geometry supplies an integer $D_r \geq 1$ such that

$$\dim P_r(Y_j) \leq D_r \quad (j \geq 1).$$

Indeed, the vertices of a Rips simplex all belong to the radius- r ball about any one of them, and these balls have uniformly bounded cardinality. The dimension bound converts the line-bundle roots into integral divisibility in K -theory.

Lemma 3.3. *For every fixed integer $r \geq 3$ and all sufficiently large j ,*

$$p_{Y,j}^* \theta_j \in 2^{j - \lfloor \log_2 D_r \rfloor} K^1(P_r(Y_j)). \quad (3.12)$$

Proof. Take j satisfying Lemma 3.2 and $2^j > D_r$, and put $z = [\mathcal{L}'_{j,r}] - 1 \in K^0(P_r(Y_j))$. The classifying map of this line bundle can be taken to land in $\mathbb{CP}^{D_r}$ by cellular approximation. The relation $([\mathcal{H}] - 1)^{D_r+1} = 0$ for the tautological line bundle $\mathcal{H}$ on $\mathbb{CP}^{D_r}$ therefore implies $z^{D_r+1} = 0$; see [1, Corollary 2.7.9]. Consequently (3.7) gives

$$[p_{Y,j}^* \mathcal{L}_j] - 1 = (1 + z)^{2^j} - 1 = \sum_{h=1}^{D_r} \binom{2^j}{h} z^h. \quad (3.13)$$

For $1 \leq h \leq D_r$, let $v_2(h)$ be the exponent of 2 dividing h . The identity

$$h \binom{2^j}{h} = 2^j \binom{2^j - 1}{h - 1}$$

shows that $\binom{2^j}{h}$ is divisible by $2^{j-v_2(h)}$, hence by $2^{j-\lfloor \log_2 D_r \rfloor}$. Every coefficient in (3.13) has this common divisor. Multiplying the resulting divisibility in K^0 by $p_{Y,j}^* e_3$ proves the claim. $\square$

We have now obtained the growing divisor over the middle. It remains to show that an equality between the classes at the two ends would force their common pairing to have this divisor.

3.5 Mayer–Vietoris detection and the direct limit

For $r \geq 3$, let

$$t_{\pm,j,r} : T_j \longrightarrow P_r(X_j)$$

denote the inclusion of the triangulated torus at height $\pm 3j$.

Lemma 3.4. *Fix an integer $r \geq 3$ and a nonzero integer n . For all sufficiently large j ,*

$$n(t_{+,j,r})_* \tau_j - n(t_{-,j,r})_* \tau_j \neq 0 \quad \text{in } K_1(P_r(X_j)). \quad (3.14)$$

Proof. In addition to the hypotheses needed for Lemma 3.3, take $j > 2r$. Cover $P_r(X_j)$ by the induced subcomplexes

$$M^+ = P_r(X_j)|_{\{t \geq -r\}}, \quad M^- = P_r(X_j)|_{\{t \leq r\}},$$

and put $I = M^+ \cap M^-$. These subcomplexes cover: a simplex not contained in either would contain vertices at heights below $-r$ and above r , contradicting its diameter bound because height is 1-Lipschitz. Their intersection is induced on the vertices at heights in $[-r, r]$.

There is a simplicial map

$$h : I \longrightarrow P_r(Y_j)$$

that forgets height. To check this, join any two vertices in a simplex of I by a graph path of length at most r in X_j . Every vertex of this path has height in $[-2r, 2r]$, which lies in the product collar since $j > 2r$. Forgetting height sends the path to a path in Y_j of no greater length. Thus the images of the simplex vertices have diameter at most r . The interpolation maps satisfy

$$p_{X,j}|_I = p_{Y,j} \circ h. \quad (3.15)$$

Let $a_{\pm} : I \hookrightarrow M^{\pm}$ and $b_{\pm} : M^{\pm} \hookrightarrow P_r(X_j)$ be the inclusions, and write $t^{\pm} : T_j \rightarrow M^{\pm}$ for the endpoint inclusions. The relevant part of the Mayer–Vietoris sequence is

$$K_1(I) \xrightarrow{((a_+)_*, -(a_-)_*)} K_1(M^+) \oplus K_1(M^-) \xrightarrow{(b_+)_* + (b_-)_*} K_1(P_r(X_j)).$$

If the difference in (3.14) were zero, exactness would give $\eta \in K_1(I)$ with

$$(a_+)_*\eta = n(t^+)_*\tau_j, \quad (a_-)_*\eta = n(t^-)_*\tau_j.$$

Pair the first equality with $(p_{X,j}|_{M^+})^*\theta_j$. Since $p_{X,j} \circ t_{+,j,r} = \text{id}_{T_j}$, naturality (3.1) and (3.3) give

$$n = \langle (p_{X,j}|_I)^*\theta_j, \eta \rangle = \langle h^*p_{Y,j}^*\theta_j, \eta \rangle. \quad (3.16)$$

By Lemma 3.3, the last integer is divisible by $2^{j - \lfloor \log_2 D_r \rfloor}$. Taking j large enough that

$$j - \lfloor \log_2 D_r \rfloor > v_2(|n|)$$

contradicts (3.16). $\square$

Proposition 3.5. *The class $\alpha \in \text{KX}_1(X)$ defined in (3.5) has infinite order. In particular, $\alpha \otimes 1 \neq 0$ in $\text{KX}_1(X) \otimes \mathbb{Q}$.*

Proof. Fix a nonzero integer n and an integer $r \geq 3$. Let α_r be the image of α_3 under the map induced by $P_3(X) \hookrightarrow P_r(X)$. For all sufficiently large j , the component separation ensures that no simplex of $P_r(X)$ meets both X_j and its complement. Thus $P_r(X_j)$ is a clopen subcomplex, and extension by zero on functions gives a $*$ -homomorphism

$$C(P_r(X_j)) \longrightarrow C_0(P_r(X)).$$

Contravariance of KK in its first variable defines restriction

$$\text{KK}_1(C_0(P_r(X)), \mathbb{C}) \longrightarrow K_1(P_r(X_j)).$$

By (3.4) and functoriality, the restriction of $n\alpha_r$ is exactly the difference in (3.14). Choosing j also large enough for Lemma 3.4 proves $n\alpha_r \neq 0$.

We have proved that for every nonzero n and every integer $r \geq 3$, the scale- r image of $n\alpha_3$ is nonzero. If $n\alpha$ vanished in the filtered direct limit defining $\text{KX}_1(X)$, it would vanish at some finite later scale; integer scales are cofinal, so this is impossible. Hence α has infinite order. The kernel of the map from any abelian group to its tensor product with $\mathbb{Q}$ is its torsion subgroup, which proves the final assertion. $\square$

4 Maximal Roe algebras and the coarse index

The infinite-order class is now constructed. It remains to show that its two end contributions have the same maximal index. The subsequent averaging and sliding arguments will use operator formulas on the standard modules. We therefore establish the norm estimates that allow those formulas to be used in the maximal completion, and then verify the required naturality of the index. These are standard features of maximal Roe theory; see [14, Section 2] and [16, Section 2]. We give the details needed here because the distinction between operator norm and maximal norm is essential to the proof.

4.1 Norm control and multipliers

Let Z be a uniformly discrete proper metric space of bounded geometry, and write

$$H_0 = \ell^2(\mathbb{N}), \quad H_Z = \ell^2(Z) \otimes H_0, \quad A_Z = C_{\max}^*(Z).$$

The algebra $\mathbb{C}[Z]$ consists of the bounded finite-propagation operators on H_Z that are locally compact for the multiplication representation of $C_0(Z)$. We denote their concrete operator norm by $\|\cdot\|_{\text{op}}$ and their maximal norm by $\|\cdot\|_{\max}$. Write $T_{xy} : H_0 \rightarrow H_0$ for the matrix entries of an operator T .

For a bounded finite-propagation operator, local compactness is equivalent to compactness of every entry T_{xy} . In one direction, compress by the characteristic functions of single vertices. Conversely, every row and column of a finite-propagation matrix has only finitely many entries. Multiplication on either side by a finitely supported function therefore gives a finite matrix of compact operators. Approximation by finitely supported functions proves the assertion for every function in $C_0(Z)$. In particular, no uniform bound on the ranks, or on finite-rank approximations to the entries, is part of this definition.

Lemma 4.1 (Fixed-propagation norm control). *For every $R \geq 0$ there is a constant $c_Z(R) < \infty$ such that*

$$\|T\|_{\text{op}} \leq \|T\|_{\max} \leq c_Z(R) \|T\|_{\text{op}} \quad (T \in \mathbb{C}[Z], \text{prop}(T) \leq R).$$

Consequently, operator-norm convergence of algebraic Roe operators with one common propagation bound implies convergence in maximal norm.

Proof. Put $N_R = \sup_{z \in Z} |B_Z(z, R)|$. The support of T , viewed as a bipartite graph with a row copy and a column copy of Z , has degree at most N_R . A greedy edge coloring partitions it into at most $2N_R - 1$ matchings. Accordingly, $T = \sum_{a=1}^{2N_R-1} T^{(a)}$, where each $T^{(a)}$ has at most one nonzero entry in each row and column, and $\|T^{(a)}\|_{\text{op}} \leq \|T\|_{\text{op}}$.

The diagonal algebra

$$\prod_{z \in Z} \mathcal{K}(H_0) = \left\{ \text{diag}(k_z) : k_z \in \mathcal{K}(H_0), \sup_z \|k_z\| < \infty \right\}$$

is a C^* -subalgebra of $\mathbb{C}[Z]$. Every $*$ -representation of $\mathbb{C}[Z]$ restricts to a contractive representation of this algebra. Since $(T^{(a)})^* T^{(a)}$ is diagonal, every such representation π satisfies

$$\|\pi(T^{(a)})\|^2 \leq \|(T^{(a)})^* T^{(a)}\|_{\text{op}} \leq \|T\|_{\text{op}}^2.$$

The upper bound follows with $c_Z(R) = 2N_R - 1$. The concrete representation on H_Z gives the lower bound. Apply the estimate to differences to obtain the convergence assertion. $\square$

This matching argument is the discrete-module form of the maximal-norm estimate in [14, Lemma 2.1]; compare [18, Lemma 1.10]. We shall also use the following stronger statement when the vertex support, rather than merely the propagation, is fixed.

Lemma 4.2 (The compact ideal). *An operator in $\mathbb{C}[Z]$ supported on $F \times F$, for a finite set $F \subset Z$, has the same maximal and concrete norms. The closure in A_Z of all such operators is an ideal $\mathcal{K}_Z$, canonically isomorphic to $\mathcal{K}(H_Z)$. In particular, $K_1(\mathcal{K}_Z) = 0$.*

Proof. The operators supported on $F \times F$ form the C^* -algebra $\mathcal{K}(\ell^2(F) \otimes H_0)$. Restriction of any representation is contractive, and the concrete representation gives the reverse inequality. These finite-vertex algebras have dense union in $\mathcal{K}(H_Z)$. Multiplying one of their elements on either side by a finite-propagation operator with compact entries gives another compact operator of finite vertex support: only a finite metric neighborhood of F can occur. Density proves the ideal assertion. $\square$

We will need finite-propagation operators whose entries are bounded but need not be compact, such as the unitaries implementing the group action. Here and below $M(A_Z)$ denotes the multiplier algebra of A_Z .

Lemma 4.3 (Finite-propagation multipliers). *Every bounded finite-propagation operator D on H_Z acts by left and right multiplication on $\mathbb{C}[Z]$, and these actions extend to a multiplier of A_Z . This assignment is a faithful $*$ -homomorphism and satisfies*

$$\|D\|_{\text{op}} \leq \|D\|_{M(A_Z)} \leq c_Z(R) \|D\|_{\text{op}} \quad \text{if } \text{prop}(D) \leq R.$$

For an operator supported on a single matching, the multiplier norm equals the concrete norm.

Proof. Products of D with elements of $\mathbb{C}[Z]$ have finite propagation and compact entries. Decompose D into matchings as in Lemma 4.1. It suffices to show that a matching operator whose entries have norm at most one acts contractively. For such an operator, D^*D is a diagonal positive contraction. The operator $W = (1 - D^*D)^{1/2}$ is bounded and diagonal. For $T \in \mathbb{C}[Z]$, all terms in

$$T^*T - (DT)^*(DT) = (WT)^*(WT)$$

belong to $\mathbb{C}[Z]$. Applying each $*$ -representation gives $\|DT\|_{\text{max}} \leq \|T\|_{\text{max}}$. Applying this to D^* and taking adjoints gives the corresponding right-multiplication bound. The algebraic compatibility identities between these left and right actions pass to the completion, so they define a multiplier. Summing the matching estimates gives the asserted upper bound.

The assignment preserves products and adjoints on the dense algebra $\mathbb{C}[Z]$. The concrete representation of A_Z on H_Z is nondegenerate, since it contains $\mathcal{K}(H_Z)$, and hence extends to the multiplier algebra. It sends the multiplier just constructed to the original operator D , proving the lower bound and faithfulness. $\square$

In particular, an operator identity involving bounded finite-propagation operators remains an identity between their multipliers. This statement does not assert that the concrete representation of the entire maximal Roe algebra is faithful. The finite-propagation control in the preceding lemmas is what will justify the specific limits below.

4.2 Maps induced by coarse maps

A map $h : Z \rightarrow W$ between uniformly discrete proper bounded-geometry spaces is coarse if inverse images of bounded sets are bounded and, for every R , there is S such that $d_Z(x, y) \leq R$ implies $d_W(h(x), h(y)) \leq S$. Choose an isometry $V_h : H_Z \rightarrow H_W$ that sends the summand over z into the summand over $h(z)$, using orthogonal ranges when several vertices have the same image. Such a choice is possible because the vertex fibers are finite and H_0 is infinite dimensional. We call V_h an exact covering isometry for h .

Proposition 4.4 (Coarse functoriality). *Conjugation by an exact covering isometry defines a $*$ -homomorphism*

$$h_A = \text{Ad}(V_h) : A_Z \longrightarrow A_W.$$

Its induced map on K -theory is independent of the covering isometry. These induced maps respect composition, and an exact covering isometry over id_Z induces the identity on $K_(A_Z)$.*

Proof. For $T \in \mathbb{C}[Z]$, coarse control bounds the propagation of $V_h T V_h^*$. Each of its vertex entries is a finite sum of compact operators, because $h^{-1}(w)$ is finite for each w . Thus conjugation is a $*$ -homomorphism $\mathbb{C}[Z] \rightarrow \mathbb{C}[W]$, and the universal property extends it contractively to the maximal completions.

For two choices V_0, V_1 , put

$$V_\lambda = \begin{pmatrix} \cos \lambda V_0 \\ \sin \lambda V_1 \end{pmatrix} : H_Z \longrightarrow H_W \oplus H_W, \quad 0 \leq \lambda \leq \pi/2.$$

Each V_λ is an isometry. Conjugation defines a homomorphism to $M_2(A_W)$: its four matrix entries have compact vertex entries and one common propagation bound for every fixed $T \in \mathbb{C}[Z]$. Moreover,

$$\|V_\lambda T V_\lambda^* - V_\nu T V_\nu^*\|_{\text{op}} \leq 2\|T\|_{\text{op}}|\lambda - \nu|.$$

Lemma 4.1, applied to the four entries, promotes this estimate to maximal-norm continuity on $\mathbb{C}[Z]$; contractivity and density give continuity on A_Z . The two endpoint homomorphisms are $\text{Ad}(V_0)$ and $\text{Ad}(V_1)$ in the two standard matrix corners, which induce the same stabilized K -theory map. Composing covering isometries proves functoriality, and comparison with the identity isometry proves the last assertion. $\square$

This is the exact-covering version of [14, Lemma 2.15]. For the maps in our construction we henceforth choose such homomorphisms and write

$$f_A : A_Y \longrightarrow A_B, \quad i_{\pm, A} : A_B \longrightarrow A_X.$$

By Proposition 4.4, all statements about their K -theory maps are independent of these choices.

4.3 The canonical odd index and naturality

The remaining index-theoretic input is compatibility with the two end inclusions.

Proposition 4.5 (Naturality for the end classes). *For the classes $\beta \in \text{KX}_1(B)$ and $\alpha = (i_+)_*\beta - (i_-)_*\beta \in \text{KX}_1(X)$ constructed above,*

$$\mu_X^{\max}(\alpha) = (i_{+, A})_* \mu_B^{\max}(\beta) - (i_{-, A})_* \mu_B^{\max}(\beta). \quad (4.1)$$

To verify this identity in the maximal completion, we first identify the Roe algebra of a Rips complex with the vertex algebra and then recall the exponential formula for its canonical odd index.

For $r > 0$, regard a point of $P_r(Z)$ as a finitely supported probability measure on Z whose support has diameter at most r . Equip the complex with the 1-Wasserstein metric induced by d_Z . Every point is within r of every vertex of its support. Moreover, a bounded set in this metric uses only vertices from one bounded subset of Z : if v is a support vertex of p , then

$$d_Z(v, z_0) \leq r + \int_Z d_Z(z, z_0) dp(z) = r + d_{W_1}(p, \delta_{z_0}).$$

That subset is finite. Consequently this metric is proper and induces the usual topology on the locally finite complex. The vertex inclusion and any selection of a support vertex are coarse inverses.

Choose a countable dense subset $E \subset P_r(Z)$ containing the vertices, and use the ample module $\ell^2(E) \otimes H_0$ with pointwise multiplication by $C_0(P_r(Z))$. Assign each point of E to one vertex of its support, assigning a vertex to itself. This partitions the module into infinite-dimensional separable blocks H_v , one for each $v \in Z$; choose identifications $H_v \cong H_0$.

Under this regrouping, the algebraic Roe algebra of $P_r(Z)$ is exactly $\mathbb{C}[Z]$. Propagation bounds in the two descriptions differ by at most $2r$. For local compactness, the points in H_v lie in the compact closed star of v , so a compactly supported continuous cutoff equal to one on that star shows that a locally compact operator has compact vertex-block entries. Conversely, a compact subset of $P_r(Z)$ meets only finitely many blocks. Compact vertex-block entries and finite propagation therefore give local compactness after multiplication by a compactly supported function, and then by any function in $C_0(P_r(Z))$. The algebraic identification extends to the maximal completions by universality. It is the coarse identification induced by the chosen support-vertex map. Pseudolocality below is always taken relative to $C_0(P_r(Z))$; the regrouping is used only for the Roe algebra and for Lemma 4.3.

We recall the cycle description of the odd index. On an ample module for $P_r(Z)$, an odd K -homology class is represented, after stabilization and addition of degenerate cycles, by a bounded self-adjoint operator F such that

$$[F, a] \text{ is compact,} \quad a(F^2 - 1) \text{ is compact} \quad (a \in C_0(P_r(Z))).$$

The first condition is pseudolocality. The use of a fixed ample module is the usual Paschke realization of locally finite K -homology; see [11].

Such a representative may be taken to have finite propagation. Indeed, let φ_v be the square roots of the barycentric coordinate functions. Their supports are compact, have diameter at most $2r$, and form a locally finite family with $\sum_v \varphi_v^2 = 1$. Set

$$F' = \sum_v \varphi_v F \varphi_v.$$

The sum converges strongly and has norm at most $\|F\|$: it is the compression of the direct sum of copies of F by the isometry $\xi \mapsto (\varphi_v \xi)_v$. It has propagation at most $2r$. Furthermore,

$$F' - F = \sum_v \varphi_v [F, \varphi_v]$$

is locally compact, since multiplication on the left by a compactly supported function leaves only finitely many compact summands; use self-adjointness for the right-hand assertion. Thus the straight-line path from F to F' consists of cycles and does not change the K -homology class. Replace F by F' .

After regrouping into vertex blocks, Lemma 4.3 makes F a self-adjoint multiplier of A_Z , and finite propagation gives $F^2 - 1 \in \mathbb{C}[Z]$. Hence

$$p = \frac{1 + F}{2} \in M(A_Z), \quad p^2 - p \in \mathbb{C}[Z], \quad u_F = \exp(2\pi i p) \in 1 + A_Z.$$

The odd maximal index is

$$\text{Ind}_{Z, \max}(F) = [u_F] \in K_1(A_Z). \quad (4.2)$$

To relate this formula to the canonical index, let $D_{\max}^*(P_r(Z))$ be the universal completion of the algebra of finite-propagation pseudolocal operators on the Rips module. It contains A_Z as an ideal, and Paschke duality identifies

$$\text{KK}_1(C_0(P_r(Z)), \mathbb{C}) \cong K_0(D_{\max}^*(P_r(Z))/A_Z), \quad [F] \longmapsto [p \bmod A_Z].$$

These assertions, including existence of the universal completion, are proved in [16, Proposition 2.22, Propositions 2.27–2.28, and Corollary 2.29]. The index is the boundary map of this extension. The multiplier action of Lemma 4.3 agrees with the inclusion on A_Z , so its exponential boundary formula is exactly (4.2). Thus this formula computes the canonical index, with its usual boundary-map sign convention. It agrees with the cycle construction in [14, Corollary 2.18]. Passing to the Rips direct limit gives $\mu_Z^{\max}$.

Proof of Proposition 4.5. Both classes are represented at Rips scale 3, where the end maps are proper continuous coarse simplicial inclusions $i_{\pm, 3} : P_3(B) \rightarrow P_3(X)$. Choose the dense-set modules compatibly: the target dense set contains the images of the source dense set, and the support-vertex choices agree on these images. The resulting module isometry V is an exact covering isometry for the end map after regrouping.

If F represents a source odd class and has finite propagation, its push-forward is represented on the target module by

$$\tilde{F} = VFV^* - (1 - VV^*).$$

The complement is a degenerate cycle. Properness ensures that pullbacks of C_0 functions lie in C_0 , so $\tilde{F}$ is pseudolocal and $\tilde{F}^2 - 1$ is locally compact. Coarse control preserves finite propagation. Writing $p = (1 + F)/2$, we have $\tilde{p} = (1 + \tilde{F})/2 = VpV^*$.

For completeness, compatibility of the exponential with conjugation holds in the maximal completion itself. The function

$$h(z) = \frac{\exp(2\pi i z) - 1}{z^2 - z}$$

extends to an entire function. Approximate h uniformly by polynomials on disks containing the spectra of the two multipliers. Since

$$\exp(2\pi i p) - 1 = (p^2 - p)h(p),$$

and $p^2 - p \in \mathbb{C}[B]$, all polynomial identities are identities on the algebraic Roe algebras, and their limits hold in maximal norm. They give

$$\exp(2\pi i \tilde{p}) - 1 = i_{\pm, A}(\exp(2\pi i p) - 1).$$

Thus the two indices commute with the end inclusion. Apply this to β_3 , subtract the two identities, and use the definition of the Rips direct-limit assembly map. Independence of the covering choices follows from Proposition 4.4. $\square$

5 A transfer in the maximal Roe algebra

The projection $f : Y \rightarrow B$ forgets a finite Cayley graph over each base vertex. We will lift operators on B to operators on Y by transporting their matrix entries between these fibers and then projecting onto the constant vectors. The resulting map supplies the analytic class that we will move from one end of X to the other.

Proposition 5.1. *For the projection $f : Y \rightarrow B$ there is a $*$ -homomorphism*

$$\Phi : A_B \longrightarrow A_Y$$

such that $(f_A\Phi)_ = \text{id}$ on $K_*(A_B)$. In particular,*

$$(f_A)_* : K_1(A_Y) \longrightarrow K_1(A_B)$$

is surjective.

Throughout this section, u, v, w denote vertices of B . Write $F_u = f^{-1}(u)$ and let

$$\xi_u = |F_u|^{-1/2} \sum_{y \in F_u} \delta_y \quad \text{in } \ell^2(F_u).$$

These vectors define an isometry

$$V : H_B \longrightarrow H_Y, \quad V(\delta_u \otimes h) = \xi_u \otimes h. \quad (5.1)$$

The diameters of the fibers are not uniformly bounded, so V itself need not be a covering isometry for a coarse map. We will obtain an element of the maximal Roe algebra from this averaging operation by using property (T).

5.1 The projection onto fiber constants

For $\gamma \in \Gamma$, let U_γ act by left multiplication by γ on every fiber, with the identity on H_0 . This is a unitary operator of propagation at most the word length of γ . Lemma 4.3 therefore gives a unitary representation

$$\Gamma \longrightarrow M(A_Y), \quad \gamma \longmapsto U_\gamma,$$

and hence a unital $*$ -homomorphism from $C_{\max}^*(\Gamma)$ into $M(A_Y)$.

We recall explicitly the consequence of property (T) that will be used; see [2]. In $C_{\max}^*(\Gamma)$ put

$$\Delta = 1 - \frac{1}{|S|} \sum_{s \in S} s.$$

For every unitary representation π and every vector η ,

$$\langle \pi(\Delta)\eta, \eta \rangle = \frac{1}{2|S|} \sum_{s \in S} \|\pi(s)\eta - \eta\|^2.$$

In particular, $0 \leq \Delta \leq 2$, and the kernel of $\pi(\Delta)$ is the invariant subspace. Property (T), applied to the fixed finite generating set S , gives $\kappa > 0$ such that every unit vector orthogonal to that subspace is moved by at least κ by some $s \in S$. Consequently, with $\varepsilon = \kappa^2/(2|S|)$,

$$\sigma_{C_{\max}^*(\Gamma)}(\Delta) \subseteq \{0\} \cup [\varepsilon, 2].$$

The continuous functional calculus thus gives a projection onto the invariant vectors in every representation, and

$$\left(1 - \frac{\Delta}{2}\right)^n$$

converges to this projection in the maximal group norm. Denote its image in $M(A_Y)$ by P , and denote the images of the displayed approximants by P_n . We have

$$\|P_n - P\|_{M(A_Y)} \longrightarrow 0. \quad (5.2)$$

Each P_n is a finite-propagation multiplier preserving every fiber.

In the concrete representation on H_Y , the invariant vectors in $\ell^2(F_u)$ are precisely the constant vectors: the map $\Gamma \rightarrow Q_{2j}$ is onto and its left action on F_u is transitive. Thus the concrete image of P is VV^* . The transfer itself will use P in the maximal multiplier algebra $M(A_Y)$.

5.2 Transporting matrix entries

For each pair u, v in the same component B_j , choose a shortest edge path from v to u , choosing the reverse path for the reversed pair. Horizontal transport along this path is a permutation

$$O_{uv} : \ell^2(F_v) \longrightarrow \ell^2(F_u).$$

Take $O_{uu} = 1$. When u, v lie in different components, instead put

$$O_{uv} = |\xi_u\rangle\langle\xi_v|.$$

In both cases,

$$O_{vu} = O_{uv}^*, \quad O_{uv}\xi_v = \xi_u, \quad O_{uv}^*\xi_u = \xi_v. \quad (5.3)$$

Moreover, O_{uv} intertwines the left Γ -actions. For horizontal transport this follows because the matchings use right multiplication; for the rank-one operators it follows because both vectors are invariant.

Given $T \in \mathbb{C}[B]$, use the decomposition $H_Y = \bigoplus_{u \in B} \ell^2(F_u) \otimes H_0$ to define

$$L(T)_{uv} = O_{uv} \otimes T_{uv}. \quad (5.4)$$

This is a linear, adjoint-preserving lift, but it need not be multiplicative. The next lemma isolates both its control and the precise failure of multiplicativity.

Lemma 5.2. *For $T \in \mathbb{C}[B]$, the operator $L(T)$ belongs to $\mathbb{C}[Y]$ and commutes with P . For $T, T' \in \mathbb{C}[B]$, the defect*

$$D = L(T)L(T') - L(TT') \quad (5.5)$$

has finite vertex support and satisfies $PD = 0$ in A_Y .

Proof. Let R be a propagation bound for T . The matrix support of T can be partitioned into finitely many matchings of the base vertices, with a number depending only on R and the bounded geometry of B . On one such matching, the blocks in (5.4) have disjoint domain and range blocks, and their norms are at most $\|T\|_{\text{op}}$. Summing over the matchings proves that $L(T)$ is bounded.

Within a component, any nonzero vertex entry of O_{uv} joins vertices connected by the chosen horizontal path. Its length is $d_B(u, v) \leq R$. Among different components, only finitely many vertex pairs (u, v) can occur at this propagation bound; their fibers are finite. The remaining entries therefore also have a finite propagation bound. Each vertex entry

of $L(T)$ is a scalar multiple of a compact operator T_{uv} . It follows that $L(T) \in \mathbb{C}[Y]$. The intertwining property of O_{uv} shows that $L(T)$ commutes with every U_γ , hence with every P_n and with P .

We next locate the defect. Let R' be a propagation bound for T' . Outside finitely many components, all entries involved in (5.5) stay in a single component B_j . For a summand with intermediate vertex v , the concatenated path from w to v to u has length at most $R + R'$, as does the chosen path from w to u . If $l_j > 2(R + R')$, these two paths have the same total seam crossing numbers. Indeed, their lifted coordinate displacements differ by a multiple of l_j and have difference of absolute value at most $2(R + R')$ in each coordinate. That multiple must be zero. Since U_j and V_j commute in Q_{2j} , horizontal transport depends only on these seam totals. It follows that

$$O_{uv}O_{vw} = O_{uw}$$

for every summand under consideration. The defect therefore vanishes on all sufficiently large components. The omitted components and any cross-component entries lie in a finite union of finite graphs, so D has finite vertex support.

Equations (5.3) give, on the concrete Hilbert spaces,

$$L(T)V = VT, \quad V^*L(T) = TV^*.$$

The concrete image of PD is consequently zero. To obtain the equality in the maximal completion, choose a finite union E of components of Y containing the vertex support of D . Every P_n preserves each fiber, so $P_n D$ has vertex support in the same finite set E . Lemma 4.2 identifies its maximal norm with its concrete operator norm. On H_Y , we have $P_n D \rightarrow 0$ in operator norm; hence $P_n D \rightarrow 0$ in maximal norm. On the other hand, (5.2) gives $P_n D \rightarrow PD$ in A_Y . Thus $PD = 0$ in A_Y . $\square$

5.3 The splitting

We can now define the transfer and compute its composition with f_A . The first task used finite vertex support. The second will use a different control: averaging can increase propagation inside a fiber, but it does not increase propagation after the fiber is projected to its base vertex.

Proof of Proposition 5.1. For $T \in \mathbb{C}[B]$, put

$$\Phi(T) = PL(T) \in A_Y.$$

Since P is a projection commuting with the lifts, Lemma 5.2 gives

$$\Phi(T)\Phi(T') = PL(T)L(T') = PL(TT') = \Phi(TT').$$

The lift preserves adjoints, so Φ is a $*$ -homomorphism on $\mathbb{C}[B]$. The universal property of the maximal norm extends it to a $*$ -homomorphism $A_B \rightarrow A_Y$.

Let $J : H_Y \rightarrow H_B$ be the covering isometry defining f_A ; it sends the summand at $y \in F_u$ isometrically into the summand at u , with orthogonal ranges for distinct y in the same fiber. For $T \in \mathbb{C}[B]$, continuity gives

$$f_A \Phi(T) = \lim_{n \rightarrow \infty} J P_n L(T) J^* \quad \text{in } A_B. \quad (5.6)$$

All operators in this limit have propagation at most $\text{prop}(T)$ on B , because P_n preserves the fibers. In concrete operator norm they converge to

$$J V V^* L(T) J^* = J V T V^* J^*.$$

The isometry $JV : H_B \rightarrow H_B$ lies over the identity of B , so $JVTJ^*J^*$ belongs to $\mathbb{C}[B]$ and has the same propagation bound. The difference from each approximant in (5.6) has this fixed propagation bound as well. Lemma 4.1 promotes the concrete convergence to maximal-norm convergence. Hence

$$f_A \Phi(T) = \text{Ad}(JV)(T). \quad (5.7)$$

By density this identity holds on A_B . Finally, Proposition 4.4 says that a covering isometry over the identity induces the identity on K -theory. Applying it to JV proves $(f_A \Phi)_* = \text{id}$. $\square$

6 Moving the analytic class to the two ends

The two endpoint maps $i_-f, i_+f : Y \rightarrow X$ are connected componentwise by the height slices of X_j . The distance between the endpoints grows with j , so a single bounded-distance homotopy does not compare their Roe-algebra maps. Instead, we place a copy of an operator at every height and compare this stack with its shift by one height. The common interior copies will cancel on K -theory.

Proposition 6.1. *The endpoint maps induce the same homomorphism*

$$(i_{+,A}f_A)_* = (i_{-,A}f_A)_* : K_1(A_Y) \longrightarrow K_1(A_X).$$

For $-3j \leq t \leq 3j$, let

$$s_{j,t} : Y_j \longrightarrow X_j, \quad (u, q) \longmapsto (u, \bar{q}, t),$$

where $\bar{q}$ is reduction from Q_{2j} to $Q_{k_j(t)}$. The map is 1-Lipschitz: it carries both Cayley edges and horizontal matchings to edges or to vertices. Consecutive slices satisfy

$$d_X(s_{j,t}(y), s_{j,t+1}(y)) = 1. \quad (6.1)$$

Choose exact covering isometries

$$D_{j,t} : H_{Y_j} \longrightarrow H_{X_j}$$

over these vertex maps. They exist because every vertex summand is a copy of the infinite-dimensional Hilbert space H_0 . At the endpoints choose them to be the restrictions of the covering isometries defining $i_{-,A}f_A$ and $i_{+,A}f_A$, respectively. For fixed j , the ranges of the $D_{j,t}$ for distinct t are orthogonal, since they lie at distinct heights.

6.1 Stacks modulo the compact operators

Let

$$q : A_X \longrightarrow A_X/\mathcal{K}_X$$

be the quotient map. If $T \in \mathbb{C}[Y]$, write $T_j = \chi_{Y_j} T \chi_{Y_j}$. The difference $T - \bigoplus_j T_j$ has finite vertex support: at a fixed propagation bound there are only finitely many cross-component entries. In particular it is compact. Define

$$S_0(T) = \bigoplus_j \sum_{t=-3j}^{3j-1} D_{j,t} T_j D_{j,t}^*, \quad (6.2)$$

$$S_1(T) = \bigoplus_j \sum_{t=-3j}^{3j-1} D_{j,t+1} T_j D_{j,t+1}^*. \quad (6.3)$$

The number of copies is unbounded, but their ranges are orthogonal. Thus $\|S_a(T)\|_{\text{op}} \leq \|T\|_{\text{op}}$, for $a = 0, 1$. If $\text{prop}(T) \leq R$, the 1-Lipschitz property gives $\text{prop}(S_a(T)) \leq R$. Every vertex entry is compact: it is a finite sum of isometric images of compact entries of T_j , since each Y_j is finite. Hence $S_a(T) \in \mathbb{C}[X]$. No bound on the ranks of these compact entries is required in the algebra used here.

The maps S_a need not be multiplicative before taking the quotient. For $T, T' \in \mathbb{C}[Y]$, however,

$$(TT')_j - T_j T'_j = \chi_{Y_j} T (1 - \chi_{Y_j}) T' \chi_{Y_j} \quad (6.4)$$

is zero except for finitely many j . Each nonzero term is compact on H_{Y_j} , since it is a matrix of compact operators over a finite vertex set. Copying it into the finitely many heights of the corresponding X_j remains compact. Orthogonality of the height ranges and (6.4) therefore show that

$$\bar{S}_a = qS_a : \mathbb{C}[Y] \longrightarrow A_X / \mathcal{K}_X$$

is a $*$ -homomorphism. By maximal universality, it extends to a $*$ -homomorphism with domain A_Y .

6.2 A continuous rotation between the stacks

We next prove that $\bar{S}_0$ and $\bar{S}_1$ induce the same map on K -theory. The rotation takes place in a doubled target module; this ensures that the ranges stay orthogonal at every intermediate angle.

For $0 \leq \lambda \leq \pi/2$, define

$$W_{\lambda,j,t} = \begin{pmatrix} \cos \lambda D_{j,t} \\ \sin \lambda D_{j,t+1} \end{pmatrix} : H_{Y_j} \longrightarrow H_{X_j} \oplus H_{X_j}, \quad -3j \leq t < 3j.$$

Each $W_{\lambda,j,t}$ is an isometry. For distinct pairs (j, t) , their ranges are orthogonal: they lie in different components or, within one component, at different heights in each target copy. In the first copy the heights are t , and in the second they are $t + 1$. Thus they assemble to an isometry

$$W_\lambda : \bigoplus_j \bigoplus_{t=-3j}^{3j-1} H_{Y_j} \longrightarrow H_X \oplus H_X.$$

On $\mathbb{C}[Y]$ define

$$F_\lambda(T) = W_\lambda \left(\bigoplus_j \bigoplus_{t=-3j}^{3j-1} T_j \right) W_\lambda^*. \quad (6.5)$$

Its diagonal matrix blocks are scalar multiples of the stacks in (6.2)–(6.3). The off-diagonal blocks are sums of $D_{j,t} T_j D_{j,t+1}^*$ and their counterparts. If $\text{prop}(T) \leq R$, then (6.1) gives propagation at most $R + 1$ for these blocks. Their vertex entries are again finite sums of compact operators. We conclude that

$$F_\lambda(T) \in M_2(\mathbb{C}[X]), \quad \text{prop}(F_\lambda(T)) \leq R + 1.$$

The same calculation as (6.4), using $W_\lambda^* W_\lambda = 1$, shows that the images of F_λ in $M_2(A_X / \mathcal{K}_X)$ are $*$ -homomorphisms on $\mathbb{C}[Y]$. Each extends to A_Y by maximal universality.

For completeness, continuity is uniform in the number of height slices. The orthogonality just established gives

$$\|W_\lambda - W_\mu\| = ((\cos \lambda - \cos \mu)^2 + (\sin \lambda - \sin \mu)^2)^{1/2} \leq |\lambda - \mu|.$$

Consequently, in concrete operator norm,

$$\|F_\lambda(T) - F_\mu(T)\|_{\text{op}} \leq 2\|T\|_{\text{op}} |\lambda - \mu|. \quad (6.6)$$

The difference has propagation at most $R + 1$, independently of the angles. Applying Lemma 4.1 to its four matrix blocks promotes (6.6) to an estimate

$$\|F_\lambda(T) - F_\mu(T)\|_{M_2(A_X)} \leq C_R \|T\|_{\text{op}} |\lambda - \mu|$$

for a constant independent of j, λ, μ . This proves point-norm continuity on the dense algebra $\mathbb{C}[Y]$. The extended homomorphisms are contractive for the maximal norm, so density proves point-norm continuity on all of A_Y .

At $\lambda = 0$ the homomorphism is $\bar{S}_0$ in the first matrix corner, and at $\lambda = \pi/2$ it is $\bar{S}_1$ in the second. The corner identifications on K -theory agree, so

$$(\bar{S}_0)_* = (\bar{S}_1)_*. \quad (6.7)$$

6.3 Canceling the interior copies

Proof of Proposition 6.1. Define the interior stack by

$$I(T) = \bigoplus_j \sum_{t=-3j+1}^{3j-1} D_{j,t} T_j D_{j,t}^*.$$

Its image $\bar{I} = qI$ is a $*$ -homomorphism on A_Y by the same compact-defect argument. Let

$$E_- = i_{-,A} f_A, \quad E_+ = i_{+,A} f_A.$$

The endpoints in the two stacks agree modulo $\mathcal{K}_X$ with E_- and E_+ , respectively. Indeed, our endpoint covering isometries agree with those defining $E_\pm$, and discarding the cross-component part of T changes its image by a compact operator. Hence

$$\bar{S}_0 = \bar{I} + qE_-, \quad \bar{S}_1 = \bar{I} + qE_+.$$

In each sum, the two homomorphisms have orthogonal ranges: one is supported on the strictly interior heights and the other on an endpoint.

Orthogonal sums induce sums on K_1 . Explicitly, if φ and ψ are orthogonal homomorphisms and u is a matrix unitary with $u - 1$ in the domain algebra, then

$$1 + (\varphi + \psi)(u - 1) = (1 + \varphi(u - 1))(1 + \psi(u - 1)),$$

and multiplication of unitaries gives addition of their K_1 -classes. Using this additivity in (6.7) and canceling $\bar{I}_*$ gives

$$q_*(E_-)_* = q_*(E_+)_*.$$

By Lemma 4.2, $\mathcal{K}_X$ is the algebra of compact operators on H_X , so $K_1(\mathcal{K}_X) = 0$. The six-term exact sequence for

$$0 \longrightarrow \mathcal{K}_X \longrightarrow A_X \xrightarrow{q} A_X/\mathcal{K}_X \longrightarrow 0$$

therefore makes $q_* : K_1(A_X) \rightarrow K_1(A_X/\mathcal{K}_X)$ injective. It follows that $(E_-)_* = (E_+)_*$, as asserted. $\square$

Proof of Theorem 1.1. The graphs X_j constructed above are finite and connected with a uniform degree bound, and their coarse disjoint union X is uniformly discrete and has bounded geometry. Proposition 3.5 gives an infinite-order class

$$\alpha = (i_+)_*\beta - (i_-)_*\beta \in \mathrm{KX}_1(X).$$

By Proposition 5.1, choose $b \in K_1(A_Y)$ satisfying

$$(f_A)_*(b) = \mu_B^{\max}(\beta).$$

Naturality of the maximal coarse index (Proposition 4.5) and Proposition 6.1 now give

$$\begin{aligned} \mu_X^{\max}(\alpha) &= (i_{+,A})_*\mu_B^{\max}(\beta) - (i_{-,A})_*\mu_B^{\max}(\beta) \\ &= (i_{+,A}f_A)_*(b) - (i_{-,A}f_A)_*(b) = 0. \end{aligned}$$

An infinite-order element remains nonzero after tensoring with $\mathbb{Q}$, so $\alpha \otimes 1$ is a nonzero element of the kernel of the rationalized maximal coarse assembly map. $\square$

References

- [1] M. F. Atiyah, *K-Theory*, notes by D. W. Anderson, W. A. Benjamin, New York–Amsterdam, 1967.
- [2] B. Bekka, P. de la Harpe, and A. Valette, *Kazhdan’s Property (T)*, New Mathematical Monographs, vol. 11, Cambridge University Press, Cambridge, 2008. doi:10.1017/CBO9780511542749.
- [3] B. Blackadar, *K-Theory for Operator Algebras*, second ed., Mathematical Sciences Research Institute Publications, vol. 5, Cambridge University Press, Cambridge, 1998.
- [4] X. Chen, Q. Wang, and G. Yu, The maximal coarse Baum–Connes conjecture for spaces which admit a fibred coarse embedding into Hilbert space, *Adv. Math.* **249** (2013), 88–130. doi:10.1016/j.aim.2013.09.003.
- [5] J. Deng, L. Guo, Q. Wang, and G. Yu, Higher index theory for spaces with an FCE-by-FCE structure, *J. Funct. Anal.* **288** (2025), no. 1, article 110679. doi:10.1016/j.jfa.2024.110679.
- [6] A. N. Dranishnikov, S. C. Ferry, and S. Weinberger, Large Riemannian manifolds which are flexible, *Ann. of Math. (2)* **157** (2003), no. 3, 919–938. doi:10.4007/annals.2003.157.919.
- [7] G. Gong, Q. Wang, and G. Yu, Geometrization of the Strong Novikov Conjecture for residually finite groups, *J. Reine Angew. Math.* **621** (2008), 159–189. doi:10.1515/CRELLE.2008.061.
- [8] L. Guo, Q. Wang, and C. Zhang, The maximal coarse Baum–Connes conjecture for spaces that admit an A-by-FCE coarse fibration structure, *J. Noncommut. Geom.*, published online December 11, 2025. doi:10.4171/JNCG/639.
- [9] N. Higson, V. Lafforgue, and G. Skandalis, Counterexamples to the Baum–Connes conjecture, *Geom. Funct. Anal.* **12** (2002), no. 2, 330–354. doi:10.1007/s00039-002-8249-5.

- [10] N. Higson and J. Roe, On the coarse Baum–Connes conjecture, in *Novikov Conjectures, Index Theorems and Rigidity, Vol. 2*, S. Ferry, A. Ranicki, and J. Rosenberg (eds.), London Mathematical Society Lecture Note Series, vol. 227, Cambridge University Press, Cambridge, 1995, pp. 227–254. doi:10.1017/CBO9780511629365.008.
- [11] N. Higson and J. Roe, *Analytic K-Homology*, Oxford Mathematical Monographs, Oxford University Press, Oxford, 2000. doi:10.1093/oso/9780198511762.001.0001.
- [12] D. A. Kazhdan, Connection of the dual space of a group with the structure of its close subgroups, *Funct. Anal. Appl.* **1** (1967), no. 1, 63–65. doi:10.1007/BF01075866.
- [13] OpenAI, A counterexample to the coarse Novikov conjecture, OpenAI Math Release preprint [OAI:A-counterexample-to-the-coarse-Novikov-conjecture-September-23-2026](#), 2026.
- [14] H. Oyono-Oyono and G. Yu, K-theory for the maximal Roe algebra of certain expanders, *J. Funct. Anal.* **257** (2009), no. 10, 3239–3292. doi:10.1016/j.jfa.2009.04.017. [arXiv:0902.2317v1](#) (numbered citations follow this version).
- [15] J. Roe, *Coarse Cohomology and Index Theory on Complete Riemannian Manifolds*, Memoirs of the American Mathematical Society, vol. 104, no. 497, American Mathematical Society, Providence, RI, 1993. doi:10.1090/memo/0497.
- [16] T. Schick and M. Seyedhosseini, On an index theorem of Chang, Weinberger and Yu, *Münster J. Math.* **14** (2021), no. 1, 123–154. doi:10.17879/59019522628.
- [17] G. Skandalis, J.-L. Tu, and G. Yu, The coarse Baum–Connes conjecture and groupoids, *Topology* **41** (2002), no. 4, 807–834. doi:10.1016/S0040-9383(01)00004-0.
- [18] J. Špakula and R. Willett, Maximal and reduced Roe algebras of coarsely embeddable spaces, *J. Reine Angew. Math.* **678** (2013), 35–68. doi:10.1515/crelle.2012.019.
- [19] R. Willett and G. Yu, Higher index theory for certain expanders and Gromov monster groups, I, *Adv. Math.* **229** (2012), no. 3, 1380–1416. doi:10.1016/j.aim.2011.10.024.
- [20] G. Yu, The Novikov conjecture for groups with finite asymptotic dimension, *Ann. of Math.* (2) **147** (1998), no. 2, 325–355. doi:10.2307/121011.
- [21] G. Yu, The coarse Baum–Connes conjecture for spaces which admit a uniform embedding into Hilbert space, *Invent. Math.* **139** (2000), no. 1, 201–240. doi:10.1007/s002229900032.